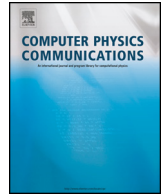




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## Computational Physics

Multi-physics coupling simulation of small mobile nuclear reactor with finite element-based models<sup>☆</sup>Xiangyue Li<sup>a</sup>, Xiaojing Liu<sup>a</sup>, Xiang Chai<sup>a</sup>, Hui He<sup>a</sup>, Bin Zhang<sup>b</sup>, Tengfei Zhang<sup>a,\*</sup><sup>a</sup> School of Mechanical Engineering, Shanghai Jiao Tong University, No. 800 Dongchuan Road, Minhang District, Shanghai, China<sup>b</sup> Nuclear power institute of China, Chengdu, China

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## ABSTRACT

This paper describes the multi-physics coupling simulation of a small transportable nuclear reactor, SIMONS, using the neutronics/thermo-mechanics coupling three-dimensional high-fidelity code platform FEMAS (FEM-based Multiphysics Analysis Software for Nuclear Reactor). FEMAS builds multi-physics field models utilizing the open-source finite element codes deal.II and FEniCS. Using coupling simulations of neutronics, thermal-conduction, and thermo-mechanics, the SIMONS micro mobile reactor is examined. For the generation of cross-sections libraries, the open-source Monte Carlo code OpenMC is utilized, and the  $(n, xn)$  correction and SPH method is developed to maintain neutron equilibrium and eliminate homogenization errors. The approaches yield neutronic solutions that are in good agreement with the reference solutions, as the  $k_{\text{eff}}$  error decreases from 5066 pcm to 7.2 pcm and the maximum relative power error decreases from 5.2% to 0.3%. With a thermal power of 240 kW, the local temperature difference of the entire reactor is 354.68 K, and the deformation rate due to thermal stress is 2.4%, resulting in a reactivity feedback of 4459.6 pcm. At the wall in contact with the outermost fuel rods, where the temperature gradient grows substantially and thermal stress rises, the monolith stress is measured to peak at 9.65 MPa. This research has paved the way for a more thorough multi-physics examination of micro nuclear reactors in the era of high-precision computation.

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## 1. Introduction

In some circumstances, a micro transportable nuclear reactor power system can function as an independent micro-grid capable of providing a long-term, high-power emergency power supply. It will reduce the loss of lives and property due to power outages, helping in communication, transportation, the military, medicine, natural disaster rescue and many other industries, offering a wide range of application prospects. At present, the United States has carried out projects to develop micro reactors such as the Prometheus [1], eVinci [2], and Holos reactors [3]. In China, Tsinghua University has designed a 200-kW gas-cooled reactor with power called IGCR-200 [4]. These reactors have been designed using a wide variety of cutting-edge technologies aiming at output powers ranging from kilowatts to megawatts.

The Small Innovative helium-xenon cooled MOBILE Nuclear power System (SIMONS) [5] proposed by Shanghai Jiao Tong Uni-

versity is a truck-transportable gas-cooled reactor designed to be contained in an international standard transportation container. SIMONS features small dimensions, high power density and inherent safety. It has a thermal capacity of 20 MW and can operate for 3,300 EFPDs without refueling. Compared to conventional reactors, SIMONS are constrained in size, weight, power-density, and lifetime, necessitating a large number of analyses to guide the reactor's optimization and operation. SIMONS is meant to function in complex environments. Multiple physical fields, including the temperature of the fuel, the temperature of the coolant, the density of the material, and other parameters, interact strongly in SIMONS. The simulation of neutronics yields the distribution of neutron flux and power. Variations in core power distribution affect the fuel temperature field, which in turn influences heat transmission from the fuel to the monolith, etc. The density of materials is altered by the variation in fuel temperature and thermal expansion of the monolith, consequently influencing macroscopic cross-sections. The current operational condition of the reactor is the consequence of the interaction and mutual impact of several physical domains. Consequently, it was determined that single-physics simulations are insufficient to represent the physical process occurring in micro reactors comprehensively.

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\* Corresponding author.

E-mail address: [zhangtengfei@sjtu.edu.cn](mailto:zhangtengfei@sjtu.edu.cn) (T. Zhang).

Numerous efforts in code and method development have been dedicated towards multi-physics coupling analysis on micro reactors. Lee et al. [6] qualitatively validated the PROTEUS/FLU-ENT/ANLHTP coupling multi-physics simulations with a 3D unit assembly problem composed of six fuel rods and seven heat pipes, which was developed based on the MegaPower compositions. In this simulation, no effect of structural change was accounted for due to the limited licensing of the ANSYS package [7]. Subsequently, Hu et al. [8] assessed the analytical capability of system analysis software (SAM) for transient modeling, analysis, and design of numerous advanced nuclear reactor systems. They coupled the MAMMOTH/SAM/MOOSE in order to simulate the reactor kinetics, the heat conduction and removal, and the thermal expansion behavior of the micro reactor. Ma et al. [9] employed RMC and ANSYS to analyze the neutronic/thermal/mechanical coupling process in heat pipe reactor, thereby predicting the thermal-mechanical feedback effect of solid reactor core. In addition, they employed the simplified point kinetics model, thermal stress model and heat pipe model to conduct the transient analysis of MegaPower [10]. Guo et al. [11] utilized RMC and OpenFOAM codes to simulate the multi-physics fields of the KRUSTY, and qualitatively examined on the thermal deformation simulation, the expansion reactivity feedback, the mesh mapping and the coupling method between two codes. Xiao et al. [13] developed MPCH, a three-dimensional (3D) high fidelity neutronics-thermo-mechanics multi-physics coupling code based on OpenMC/Nektar++/SfePy. The multi-physics field distributions of KRUSTY were accurately predicted, highlighting the significance of multi-physics coupling in evaluating the physical properties of small advanced reactors. Matthews et al. [12] used the DireWolf code to undertake a tightly coupling analysis on the coupling of thermomechanical-neutronics physics of Empire reactor using Griffin/Bison/Sockeye under MOOSE framework. This platform has offered an advanced tool for multi-physics analysis to aid in reactor design, optimization, and safety assessment. Nevertheless, it should be noted that Griffin + MOOSE is subject to export controls and has primarily been validated for heat pipe reactors. Consequently, there arises a necessity to develop a novel code system for conducting neutronics/thermo-mechanics multi-physics coupling research on the SIMONS platform.

Currently, the majority of neutronics/thermo-mechanics multi-physics coupling researches focus on heat pipe reactors. For neutronics calculation, the Monte Carlo code are preferred, but it cannot be employed to effectively conduct transient modeling and analysis. Consequently, the more practical method for analysis of various advanced nuclear reactor systems is the deterministic method. For thermo-mechanics calculation, the use of permission will eventually restrict the use of commercial software such as ANSYS, which sets obstacles for secondary development. To solve these major development barriers of multi-physics coupling researches for SIMONS, this research developed FEMAS (FEM-based Multiphysics Analysis Software for Nuclear Reactor), a coupling 3D high-fidelity computer platform for neutronics/thermo-mechanics analysis. For neutronics calculation, it adopts a high-fidelity 3D multi-group neutron diffusion/transport model with unstructured grids to estimate the neutron flux field and heat release distribution of the nuclear reactor core. For thermo-mechanics calculation, it can establish a 3D thermal diffusion model and stress analysis model to accurately obtain the temperature field and deformation. This multi-physics platform provides precise assessments of neutron physics, heat conduction, and deformation, consequently providing technical support for the fundamental design of the mobile nuclear reactor power system.

The remaining sections of the paper are structured as follows. Section 2 provides an overview of the multi-physics coupling strategy of SIMONS. Section 2.1 discusses the neutronics method, and

Section 2.2 illustrates the thermo-mechanics method. For each physics method, we provide the equations that are being resolved, as well as the optimization strategies employed in these methods. Section 2.3 outlines the framework for multi-physics implementation. Section 3 provides an overview of the simplified SIMONS core and the detailed models employed in this study. Section 4 summarizes the application of FEMAS for the simplified SIMONS, including an analysis of correction method for neutronic cross-section library generation and multi-physics coupling results. Section 5 provides a conclusion and summary.

## 2. Numerical methods

For the small nuclear reactor, three physical processes are of significant interest: neutronics, heat conduction, and thermal expansion. In this study, neutronics, thermo-conduction, and thermo-mechanics reactions are modeled and solved using the OpenMC [14], deal.II [15][16], and FEniCS [17][18] codes, and the physical fields (heat release, temperatures, and densities) are transferred between these codes for accurate reactor modeling. This section presents the coupled physical methodologies required to comprehend the thermo-mechanics performance on a multi-physics scale for a micro nuclear reactor design.

### 2.1. Neutronics

#### 2.1.1. Weak form of multigroup neutron diffusion

The neutron diffusion equation is the governing equation that states that the balance of neutrons within a system, in which the neutron current is proportional to the gradient of the neutron flux by means of a diffusion coefficient. Considering the multigroup diffusion equation for all  $g \in \{1, \dots, G\}$  with homogenized material properties, for all  $r$  in the spatial domain  $D$  [21]:

$$-\nabla \cdot (D_g(r) \nabla \phi_g) + \Sigma_{r,g}(r) \phi_g = \frac{\chi_g(r)}{k_{\text{eff}}} \sum_{g'=1}^G \nu \Sigma_{f,g'}(r) \phi_{g'} + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g}(r) \phi_{g'} \quad (1)$$

In Eq. (1),  $G$  is the total number of energy groups,  $\phi_g$  is the scalar flux,  $D_g$  is the diffusion coefficient, and  $\Sigma_{r,g}$ ,  $\Sigma_{f,g'}$  and  $\Sigma_{s,g' \rightarrow g}$  are the macroscopic removal, fission and scattering cross-sections, respectively. In addition,  $k_{\text{eff}}$  is the multiplication factor (or eigenvalue),  $\nu$  is the average number of neutrons released per fission,  $\chi_g$  is the fission spectrum.

The neutronic method is based on the finite element method (FEM) [19] in order to support a high-fidelity simulation of nuclear reactors. The weak form of the multigroup diffusion equation is derived from the FEM by multiplying Eq. (1) for all  $g$  by a test function  $\Phi^*$  from the chosen finite element space  $V$  and integrating the divergence terms by parts to introduce boundary conditions. Therefore, we then find  $\Phi^* = (\phi_1^*, \dots, \phi_G^*)^T \in V^G$  that we can obtain:

$$\begin{aligned} & \sum_{g=1}^G \left[ \int_D (D_g \nabla \phi_g \nabla \phi_g^* + \Sigma_{r,g} \phi_g \phi_g^*) dx + \int_{\partial D} \frac{1}{2} \phi_g \phi_g^* ds \right] \\ &= \sum_{g=1}^G \left( \frac{\chi_g}{k_{\text{eff}}} \sum_{g'=1}^G \int_D \nu \Sigma_{f,g'} \phi_{g'} \phi_g^* dx + \sum_{g' \neq g}^G \int_D \Sigma_{s,g' \rightarrow g} \phi_{g'} \phi_g^* dx \right) \end{aligned} \quad (2)$$

The result of  $\int_{\partial D} \frac{1}{2} \phi_g \phi_g^* ds$  depends on the boundary conditions. The Albedo boundary conditions are considered to be:

$$\nabla \phi_g(\mathbf{r}) \cdot \mathbf{n} + \frac{1}{D_g} \frac{1-\beta}{2} \phi_g(\mathbf{r}) = 0, \quad \mathbf{r} \in \partial D \quad (3)$$

where  $\mathbf{n}$  is the outward normal unit vector on the boundary,  $\beta$  is the albedo factor. This value can be chosen as either  $\beta = 0$  (vacuum boundary condition), or  $\beta = 1$  (zero-current boundary condition).

### 2.1.2. (n, xn) reaction correction

High order (n, xn) reactions do not play a major role in conventional reactors because of their high threshold energy. In SIMONS, along with the other light elements, beryllium is utilized as the neutron moderator due to its excellent moderating capabilities and small neutron-absorption cross-section. However,  $\text{Be}^9$  (n, xn) reactions, which result in neutron multiplication effects, necessitate the modification of beryllium cross-sections in order to maintain neutron equilibrium.

The multigroup diffusion equation with considering (n, xn) reactions is:

$$\begin{aligned} & -\nabla \cdot (D_g \nabla \phi_g) + \Sigma_{r,g} \phi_g \\ &= \sum_{h=1}^G \Sigma_{n1n}^{h \rightarrow g} \phi_h + 2 \sum_{h=1}^G \Sigma_{n2n}^{h \rightarrow g} \phi_h \\ &+ 3 \sum_{h=1}^G \Sigma_{n3n}^{h \rightarrow g} \phi_h + \frac{\chi_g}{k_{\text{eff}}} \sum_{h=1}^G \nu \Sigma_{f,h} \phi_h \end{aligned} \quad (4)$$

with  $\Sigma_{r,g} = \Sigma_{a,g} + \Sigma_{n1n,g} + \Sigma_{n2n,g} + \Sigma_{n3n,g} + \Sigma_{n4n,g}$ , where  $g$  and  $h$  are the energy group indices,  $\Sigma_a$  is the absorption cross-section,  $\Sigma_{n1n}$  is (n, 1n) or scattering cross-section,  $\Sigma_{n2n}$ ,  $\Sigma_{n3n}$ , and  $\Sigma_{n4n}$  are (n, 2n), (n, 3n), and (n, 4n) cross-sections, respectively.

The scattering and absorption cross-sections can be adjusted for computational convenience while keeping the form of the multigroup diffusion equation unchanged:

$$\tilde{\Sigma}_{s,g} \equiv \Sigma_{n1n,g} + 2 \Sigma_{n2n,g} + 3 \Sigma_{n3n,g} \quad (5)$$

$$\tilde{\Sigma}_s^{h \rightarrow g} \equiv \Sigma_{n1n}^{h \rightarrow g} + 2 \Sigma_{n2n}^{h \rightarrow g} + 3 \Sigma_{n3n}^{h \rightarrow g} \quad (6)$$

$$\tilde{\Sigma}_{a,g} \equiv \Sigma_{a,g} - \Sigma_{n2n,g} - 2 \Sigma_{n3n,g} \quad (7)$$

Utilizing the forgoing adjusted cross-sections, the multigroup diffusion equation can be written as:

$$\begin{aligned} & -\nabla \cdot (D_g \nabla \phi_g) + \tilde{\Sigma}_{r,g} \phi_g \\ &= \sum_{g' \neq g}^G \tilde{\Sigma}_{s,g' \rightarrow g} \phi_{g'} + \frac{\chi_g}{k_{\text{eff}}} \sum_{g'=1}^G \nu \Sigma_{f,g'} \phi_{g'} \end{aligned} \quad (8)$$

with  $\tilde{\Sigma}_{r,g} = \tilde{\Sigma}_{a,g} + \tilde{\Sigma}_{s,g}$ .

### 2.1.3. SPH method

The cross-sections used in neutronic calculations are obtained by homogenizing the space and energy variables, which introduces inaccuracies into the calculation results. To reduce homogenization errors, the Super-homogenization (SPH) method [20] is utilized to correct the homogenized cross-sections. The objective of the SPH model is to preserve the reaction rates, the leakage rate and the eigenvalue within homogeneous regions with respect to a reference heterogeneous problem [20]. For each macro region  $m \in \{1, \dots, M\}$  and energy group  $g$ , a unique SPH factor is employed to correct the cross-sections (i.e., removal, fission, and scattering):

$$\Sigma_{m,g} = \mu_{m,g} \Sigma_{m,g}^{\text{ref}} \quad (9)$$

According to the principle of SPH method, the average reaction rate  $\tau_{m,g}$  should be preserved for all  $(m, g)$ :

$$\tau_{m,g} = \mu_{m,g} \Sigma_{m,g}^{\text{ref}} \phi_{m,g} = \Sigma_{m,g}^{\text{ref}} \phi_{m,g}^{\text{ref}} = \tau_{m,g}^{\text{ref}} \quad (10)$$

where  $\phi_{m,g}^{\text{ref}}$  is the reference heterogeneous neutron flux in macro region  $m$  and group  $g$ ,  $\phi_{m,g}$  is the homogeneous neutron flux in macro region  $m$  and group  $g$ .

Therefore, the SPH factors can be calculated with:

$$\mu_{m,g} \equiv \frac{\phi_{m,g}^{\text{ref}}}{\phi_{m,g}} \quad (11)$$

However, the value of flux obtained in both heterogeneous and homogeneous calculations are meaningless since the neutron equation is an eigenvalue equation. To preserve the flux-volume averaged flux for each energy group  $g$  between the homogeneous and heterogeneous calculations, it is necessary to include a scaling factor  $\eta$  to limit the problem to a single set of solutions:

$$\eta_g \equiv \frac{\bar{\phi}_g}{\phi_g^{\text{ref}}} = \frac{\int_D \phi_g dx}{\int_D \phi_g^{\text{ref}} dx} = \frac{\sum_m V_m \phi_g^m}{\sum_m V_m \phi_g^{\text{ref},m}} \quad (12)$$

where  $\bar{\phi}$  is the average of  $\phi$  over the spatial domain  $D$ .

Apply this scaling factor  $\eta$ , the updated SPH factors and cross-sections are written as follows:

$$\tilde{\mu}_{m,g} \equiv \frac{\phi_{m,g}^{\text{ref}}}{\phi_{m,g}} \eta_g \quad (13)$$

$$\tilde{\Sigma}_{m,g} \equiv \tilde{\mu}_{m,g} \Sigma_{m,g}^{\text{ref}} \quad (14)$$

Therefore, the weak form of SPH-corrected multigroup diffusion equation is then written as:

$$\begin{aligned} & \sum_{g=1}^G \sum_{m=1}^M \int_D \left( \tilde{\mu}_{m,g} D_{m,g}^{\text{ref}} \nabla \phi_g \nabla \phi_g^* + \tilde{\mu}_{m,g} \Sigma_{r,m,g}^{\text{ref}} \phi_g \phi_g^* \right) dx \\ &+ \sum_{g=1}^G \int_{\partial D} \frac{1}{2} \phi_g \phi_g^* ds \\ &= \sum_{m=1}^M \sum_{g=1}^G \left[ \frac{\chi_{m,g}}{k_{\text{eff}}} \sum_{g'=1}^G \int_D \left( \tilde{\mu}_{m,g'} \nu \Sigma_{f,m,g'}^{\text{ref}} \phi_{g'} \phi_{g'}^* \right) dx \right. \\ &\left. + \sum_{g' \neq g}^G \int_D \left( \tilde{\mu}_{m,g'} \Sigma_{s,m,g' \rightarrow g}^{\text{ref}} \phi_{g'} \phi_g^* \right) dx \right] \end{aligned} \quad (15)$$

These reference fluxes  $\phi_{m,g}^{\text{ref}}$  and cross-section sets  $\Sigma_{m,g}^{\text{ref}}$  are generated by Monte Carlo code OpenMC for different combinations of perturbations for the selected cross-section variables (fuel temperature, monolith temperature, etc.).

Fig. 1 depicts the SPH iteration process, and the steps of the SPH iterative procedure are as follows:

- (1) Obtain the few-group cross-sections  $\Sigma_{m,g}^{\text{ref}}$  and average heterogeneous flux  $\phi_{m,g}^{\text{ref}}$  for each macro region  $m$  and group  $g$  from the OpenMC reference calculation;
- (2) Initialize the SPH factors to 1;
- (3) Multiply the SPH factors to correct the few-group cross-sections according to Eq. (14);
- (4) Using the corrected cross-sections, construct the source term of Eq. (2) with the average homogeneous flux solution  $\phi_{m,g}^k$ ;
- (5) Perform the deal.II calculations, solve for  $\phi_{m,g}^{k+1}$  by inverting the left side of Eq. (2);

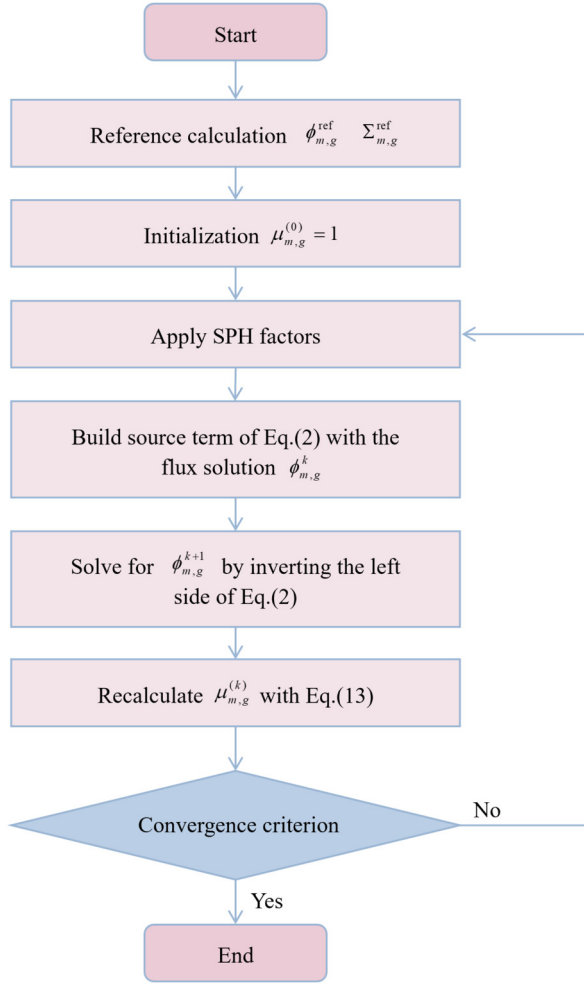


Fig. 1. The SPH iterative process.

- (6) Evaluate the SPH factor  $\mu_{m,g}^{(k)}$  using Eq. (13);  
 (7) Repeat steps (3)–(6), till the convergence criterion is satisfied for macro region  $m$  and group  $g$ , or the execution reaches the maximum number of iteration steps. For the convergence criterion of SPH factors,  $\epsilon_\mu$  is taken to be  $10^{-4}$ :

$$\max_{m,g} \frac{|\mu_{m,g}^{(k+1)} - \mu_{m,g}^{(k)}|}{\mu_{m,g}^{(k+1)}} < \epsilon_\mu \quad (16)$$

## 2.2. Thermo-mechanics

For the thermal conduction method, the differential equation of thermal conduction equation in Cartesian coordinates is:

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (\lambda \nabla T) + q_v \quad (17)$$

where  $\lambda$  is the thermal conductivity and  $q_v$  is the volume heat release rate. Changes in thermal conductivity with time and temperature is not considered in this study. Hence, Eq. (17) can be cast into the variational form:

$$-\int_{\partial D} \mathbf{n} \cdot \nabla T v ds + \int_D \nabla T \cdot \nabla v dx = \int_D \frac{q_v}{\lambda} v dx \quad (18)$$

where  $\mathbf{n}$  is the normal vector to the boundary  $\partial D$ .

The thermo-mechanics model assumes an isotropic body with ideal elasticity. The relationship between the deformation component (strain:  $\epsilon$ ) and the stress component (stress:  $\sigma$ ) is [22]:

$$\sigma = \lambda \text{tr}(\epsilon) I + 2\mu \epsilon \quad (19)$$

$$\epsilon_{ij} = \frac{1}{2\mu} \left( \sigma_{ij} - \frac{\lambda}{3\lambda + 2\mu} \sigma_{kk} \delta_{ij} \right) \quad (20)$$

where  $\lambda$  and  $\mu$  are Lamé constants, and  $I$  denotes the identity tensor. The transformation relationship between them and elastic modulus  $E$ .

The equation that governs the stress is:

$$-\nabla \cdot \sigma = \mathbf{f} \quad (21)$$

where  $\mathbf{f}$  is the external force per unit volume in the entire system.

By multiplying the preceding equation with a vectorized test function  $v$  and integrating over the domain  $D$ , the weak formulation of the linear elasticity equation can be obtained using the Green's theorem as follows:

$$-\int_{\partial D} \sigma(u) \cdot \mathbf{n} \cdot v ds + \int_D \sigma(u) : \nabla v dx = \int_D \mathbf{f} \cdot v dx \quad (22)$$

where  $\mathbf{n}$  is the normal vector to the boundary  $\partial D$  and: denotes the double dot product of tensors.

Since  $\sigma$  is symmetric, this equation can be equivalently written as:

$$-\int_{\partial D} \sigma(u) \cdot \mathbf{n} \cdot v ds + \int_D \sigma(u) : \epsilon(v) dx = \int_D \mathbf{f} \cdot v dx \quad (23)$$

where  $\epsilon(u) = \frac{1}{2} (\nabla u + \nabla u^T)$ .

Variations in temperature cause deformations in elastically unconstrained solids. Consequently, the global strain field is decided by mechanical and thermal processes. In terms of the linear small deformation model, the total strain can be decomposed into the sum of its mechanical and thermal components as follows:

$$\epsilon_{ij} = \epsilon_{ij}^{(M)} + \epsilon_{ij}^{(T)} \quad (24)$$

For isotropic materials,

$$\epsilon_{ij}^{(T)} = \alpha (T - T_0) \delta_{ij} \quad (25)$$

where  $\alpha$  is the linear thermal expansion coefficient of the material.

Therefore, the expression for the total strain is:

$$\epsilon_{ij} = \frac{1+\nu}{E} \sigma_{ij} - \frac{\nu}{E} \sigma_{kk} \delta_{ij} + \alpha (T - T_0) \delta_{ij} \quad (26)$$

Thus, the linearized thermo-mechanical constitutive equation is given by:

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij} - (3\lambda + 2\mu) \alpha (T - T_0) \delta_{ij} \quad (27)$$

Similar as the elasticity equation, the weak formulation of linear thermo-mechanics can be obtained as:

$$-\int_{\partial D} \sigma(u) \cdot \mathbf{n} \cdot v ds + \int_D \sigma(u) : \epsilon(v) dx = \int_D \mathbf{f} \cdot v dx \quad (28)$$

where  $\sigma = (\lambda \text{tr}(\epsilon) - (3\lambda + 2\mu) \alpha (T - T_0)) I + 2\mu \epsilon$ .



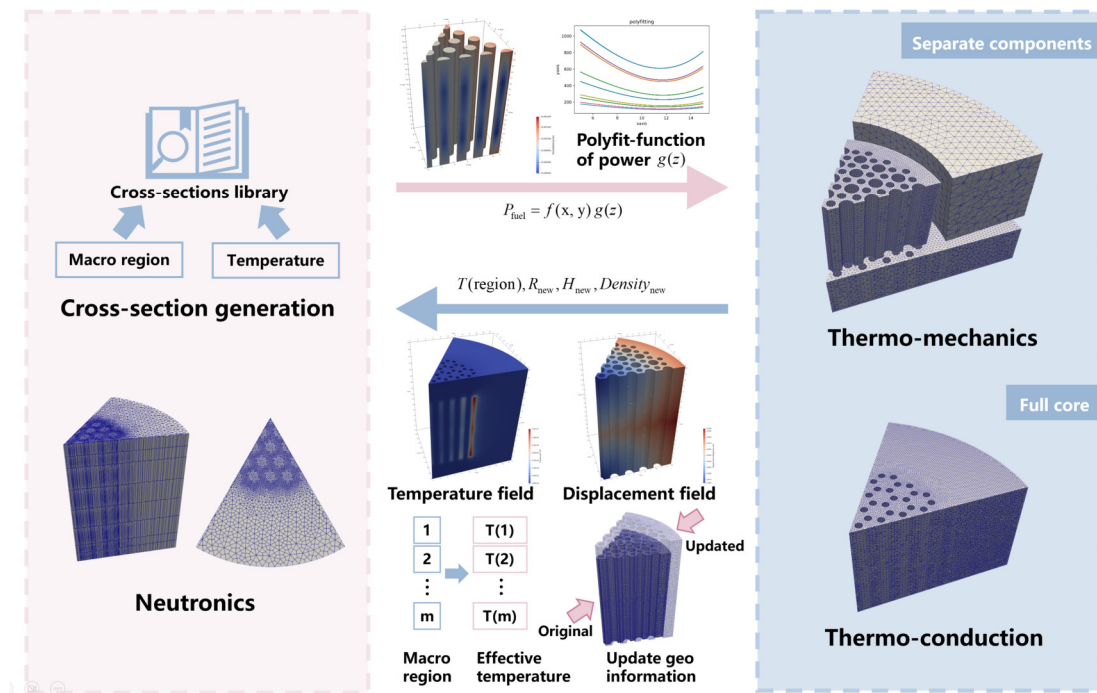


Fig. 2. A schematic view of the coupling framework.

### 2.3. Multi-physics coupling scheme

Based on open-source platforms OpenMC, deal.II and FEniCS, FEMAS integrates the neutronic and thermo-mechanics methods to solve multi-physics coupling problem. Fig. 2 displays the coupling procedure of SIMONS. On the one hand, the neutronics calculation provides power distribution for thermo-mechanics calculation which is the fitting polynomial of axial power of each fuel rod. On the other hand, the thermal-conduction calculation provides the effective temperature  $\bar{T} = \int_D T dx / \int_D dx$  of different regions to update the cross-sections used in neutronic calculation. The thermo-mechanics calculation provides structural displacement to update meshes of the overall calculation and cross-sections changed by density-decrease. By utilizing the aforementioned multi-physics processes, it is possible to determine the characteristics of the reactor core under neutronics/thermo-mechanics coupling conditions.

The coupling framework in this paper is based on the external coupling method using Picard iteration. During each iteration, deal.II firstly calculates the power distribution according to the density, temperature and geometric deformation calculated in the previous iteration. Then, FEniCS will be activated to update the density distribution, temperature distribution and the geometric deformation results. The simulation is terminated according to the criterion that

$$\varepsilon_\varsigma = \frac{|\varsigma^{n+1} - \varsigma^n|}{\varsigma^{n+1}} \quad (29)$$

where  $\varsigma$  is variable of interest (in this paper,  $k_{\text{eff}}$ , radius of monolith and effective temperature), and  $n$  is the iteration step. When  $\varepsilon$  reaches a pre-set tolerance, the algorithm is stopped.

The detailed process of coupling iteration proceeds as follows:

(1) Use OpenMC to perform neutron transport simulation and generate a cross-section library, and use Gmsh to create simulation meshes;

(2) Use deal.II to solve the multi-group neutron diffusion equation based on the established temperature distribution and geo-

metric parameters in the cold state in order to obtain the spatial power distribution;

(3) Using FEniCS to solve the thermal diffusion equation, the temperature distribution is determined based on the boundary conditions of the helium-xenon cooling channel and the spatial distribution of power;

(4) The structural displacement is obtained by solving the thermo-mechanics equation using FEniCS in conjunction with the temperature field and mechanical boundary conditions;

(5) Update the model's shape, density, and cross-sections. Please be informed that the cross-sections are interpolated using temperature and the rate of density change as reference parameters. This approach ensures a closer correspondence between the interpolated cross-sections and the underlying physics;

(6) deal.II then recalculates the neutron diffusion equation. The preceding steps are repeated until specific physical quantities satisfy convergence criteria or the maximum number of iteration steps is reached.

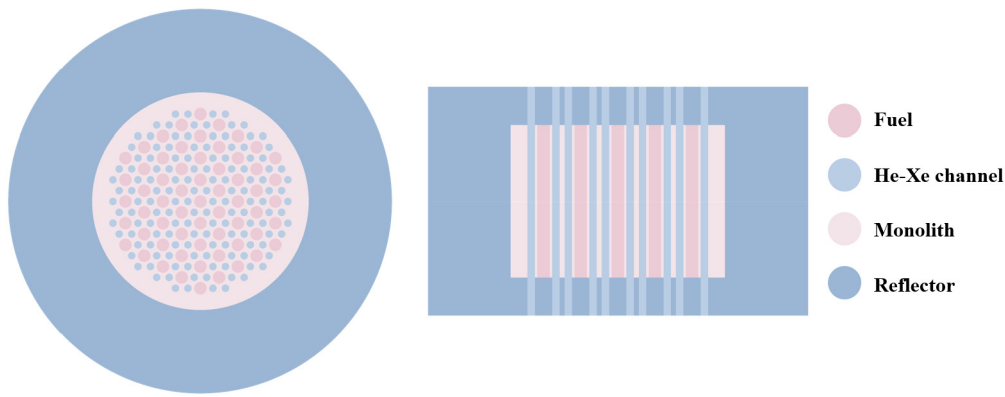
## 3. Calculation models

### 3.1. Description of simplified SIMONS

As a test of the coupling platform, a simplified model of the SIMONS core with a thermal power of 0.24 MW is employed to reduce the computational cost. Uranium-Carbide (UC) fuel with a low enrichment of 19.75% and the helium-xenon mixture coolant with a volume ratio of 88:12 are employed. The scaled core has a prismatic design and is composed of graphite monolith. In order to accommodate fuel rods and coolant, the monolith is perforated with a series of hexagonal holes. There are a total of 61 fuel rods and 138 coolant channels. As the reflector, beryllium is used. The core has an equivalent height of 30 cm and a radius of 23 cm. The axial-radial schematic diagram of the core can be seen in Fig. 3.

The geometric parameters of the simplified SIMONS are shown in Table 1.

Utilizing the symmetry of the reactor to save computing expense, just half of the reactor is modeled in the calculation model.



**Fig. 3.** A schematic diagram of the SIMONS core. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

**Table 1**

Design parameters of the simplified SIMONS core.

| Parameters                 | Value | Unit |
|----------------------------|-------|------|
| Fuel diameter              | 1.5   | cm   |
| Coolant channel diameter   | 0.9   | cm   |
| Fuel pitch                 | 1.5   | cm   |
| Number of fuel rods        | 61    | /    |
| Number of Coolant channels | 138   | /    |
| Fuel height                | 10    | cm   |
| Monolith radius            | 13    | cm   |
| Radial reflector width     | 10    | cm   |
| Axial reflector height     | 5     | cm   |

**Table 2**

Material properties used in the multi-physics coupling calculations.

| Description                               | Value                 | Unit      |
|-------------------------------------------|-----------------------|-----------|
| Young's modulus of monolith               | 11.5                  | GPa       |
| Poisson's ratio of monolith               | 0.15                  | /         |
| Linear expansion coefficient of monolith  | $8 \times 10^{-6}$    | /K        |
| Young's modulus of reflector              | 287                   | GPa       |
| Poisson's ratio of reflector              | 0.08                  | /         |
| Linear expansion coefficient of reflector | $11.5 \times 10^{-6}$ | /K        |
| Thermal conductivity of monolith          | 129                   | W/(m · K) |
| Thermal conductivity of fuel              | 25.3                  | W/(m · K) |
| Thermal conductivity of reflector         | 73                    | W/(m · K) |

Table 2 lists the properties of all the materials used in this multi-physics simulation.

### 3.2. Neutronic model of SIMONS

#### 3.2.1. Generation of cross-section library

The generation of the cross-section library should take into consideration the variables necessary to capture the most relevant feedback mechanisms (i.e., the ones that lead to significant changes in cross-sections, and thus in reactivity). For steady-state simulation, the reactivity feedback is predominantly governed by the fuel and monolith temperatures. The cross-section library is therefore parameterized with a functional formulation:

$$\Sigma_{m,g}(\mathbf{r}) = f(T_{\text{fuel}}, T_{\text{cool}}) \quad (30)$$

where  $f$  is approximated by a piece-wise linear function using the local parameter values at each element of the mesh from other physics models. The temperature grids selected for every parameter are:

- 1)  $T_{\text{fuel}} = [298, 900, 1200, 1400]\text{K}$
- 2)  $T_{\text{cool}} = [298, 800, 1000, 1200]\text{K}$

No changes in density are applied when varying the fuel temperature during the cross-section generation process. Changes in the monolith density, e.g., due to thermal expansion, will be accounted for by the mesh deformation capability of FEMAS.

Due to the intricacy of the reactor geometry and the significant neutron leakage from the core of SIMONS, the open-source Monte Carlo code OpenMC is utilized. During computations, OpenMC uses 500,000 particles for 300 batches (100 inactive batches) to generate two-group homogenized cross-sections with the following group structure:

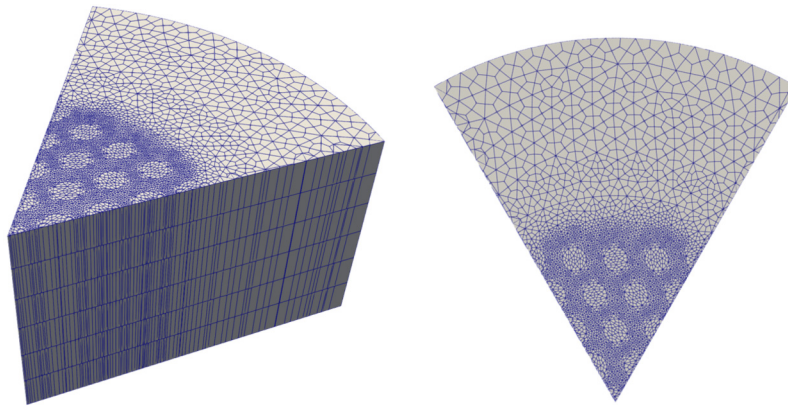
- 1) Group-1 (0.625 eV – 20 MeV)
- 2) Group-2 (0 eV – 0.625 eV)

It is well recognized that the two-group energy structure may not be adequate to capture the energy dependency of the cross sections. As a result of graphite's moderating effect, the reactor has a thermal neutron spectrum, making the use of a group condensation scheme efficient and rational. In addition, as will be demonstrated in the following sections, the combination of a whole-core cross-section generation model with the SPH factors will ensure that the reaction rates between the core diffusion calculations and the Monte Carlo calculations are in good agreement. We therefore see the two-group energy structure as a valid compromise between precision and efficiency.

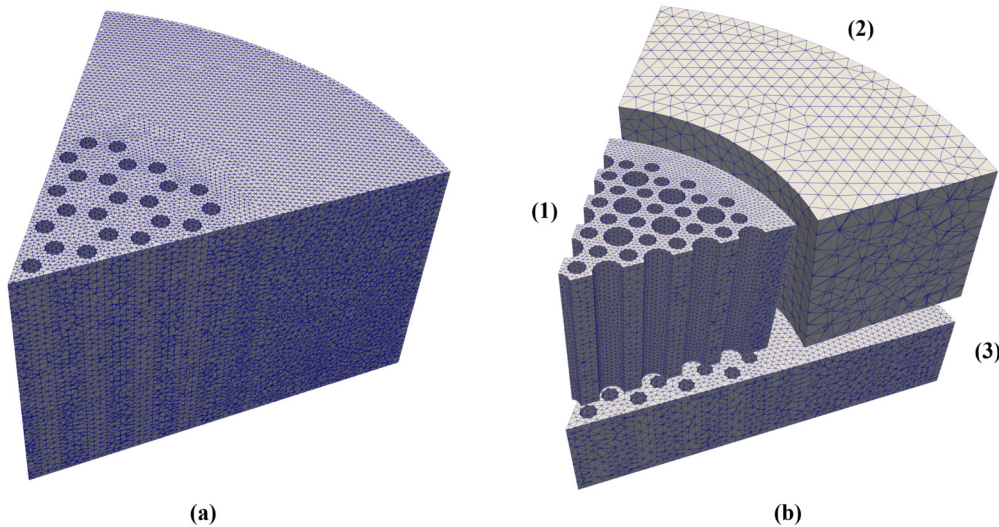
The SPH factors are computed by OpenMC and deal.II at the selected state points and saved in the cross-section library. The resulting data set consisting of multigroup data (cross-sections and SPH factors) can subsequently be utilized in a multi-physics simulation. To accurately account for the multiplication effects of (n, xn) reactions stated in Section 2.1.2, however, the scattering, absorption, and total cross-sections generated by OpenMC must be modified using (n, xn) correction prior to the commencement of the SPH iteration.

#### 3.2.2. Neutronic model

For the neutronics calculations, 3D unstructured meshes are used to explicitly describe all the rods and holes in the coupling calculations. The mesh used in the neutronic computation is depicted in Fig. 4. The cross-sections of each macro region are dependent on temperature and density. During the neutronic calculations, there needs two steps for cross-section update. First, the cross-sections for each macro zone are updated from the cross-section table by linear interpolation based on the effective temperature determined by initialization or thermal conduction calculation results. Second, all cross-sections are corrected based on the density change rate determined by thermo-mechanics calculations in each macro region. Although when considering deformation during the generation of cross-sections, the most precise approach



**Fig. 4.** The 3D finite element mesh used in the neutronic simulation.



**Fig. 5.** The finite element meshes used in thermo-mechanics simulation. (a) Full core; (b) Separate components (Monolith, radial reflector and axial reflector).

entails invoking OpenMC calculations with updated geometries at each iterative step. However, this approach poses a significant computational burden due to the repeated OpenMC calculations required. Nevertheless, during the operation of the SIMONS reactor, the magnitude of deformation is small. To mitigate computational costs, instead of employing an on-the-fly cross-section generation scheme, we assume that the influence of geometric changes on the energy spectrum (microscopic cross-sections) is minimal, and that the primary impact of deformation on neutronics manifests as variations in nuclide density. Consequently, we approximate the influence of deformation on macroscopic cross-sections in OpenMC calculations by proportionally adjusting them based on the change ratio of nuclide density.

### 3.3. Thermo-mechanics model

The mesh used in thermo-mechanics calculations is depicted in Fig. 5. The thermal conductivities of the fuel and the monolith are assumed to be constant when calculating heat diffusion. The current practice treats coolant channels as Dirichlet boundaries with the temperature of 1000 K, and future research will investigate the heat transfer of the coolant flow in the channel. On the basis of the temperature distribution obtained by solving the thermal diffusion equation, it is possible to simulate the thermal expansion process. In the thermo-mechanics model, the thermal conduction calculation uses the complete core model (illustrated in Fig. 5 (a)), which includes the fuel, monolith, radial and axial reflector. Dur-

ing the operation of the SIMONS reactor, the distinct properties of the monolith (graphite) and reflector (Beryllium) materials will cause them to separate due to disparate thermal expansion effects. Subsequent computational simulations have also confirmed this observation. Therefore, we divide the thermo-mechanical calculation models into three distinct components, as numbered in Fig. 5 (b). The temperature distribution of each component is derived from the mapping of the thermal conduction calculation results of the entire reactor core. These temperature distributions are subsequently used to perform thermal expansion calculations for each individual component.

## 4. Results

### 4.1. Neutronic results with cross-section correction method

On the basis of this simplified SIMONS reactor, three calculation models with varying degrees of precision in describing reactor core properties are proposed in this work. The details of these three instances are provided below.

Pin-cell model (shown in Fig. 6 (a)): employs a 3D unit-cell model of a single fuel rod surrounded by six coolant pins with reflective boundary conditions. Even though it does not precisely replicate the behavior of the full-core, this unit-cell problem is valuable for solving multi-physics problems at a reduced computational cost.



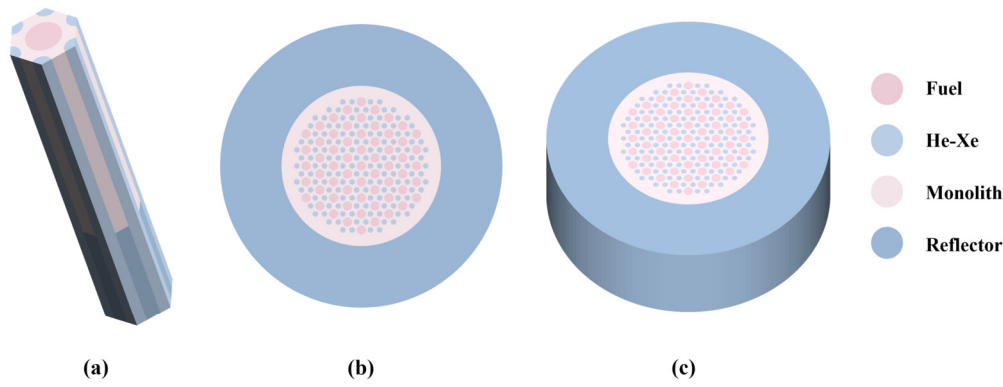


Fig. 6. The Monte Carlo models used in cross-section generation.

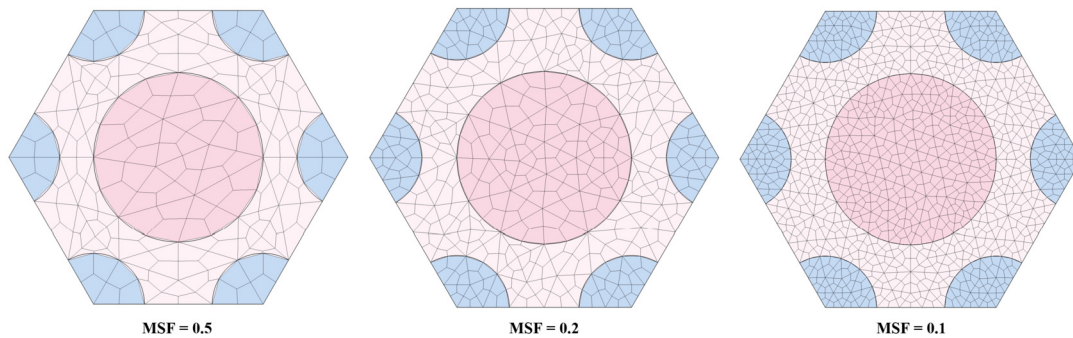


Fig. 7. Finite element meshes using different mesh size factors (MSFs).

2D model (shown in Fig. 6 (b)): employs a 2D radial model which depicts the horizontal cut of the simplified SIMONS core, enabling a more precise description of the neutron cross-sections of each fuel rod, coolant, monolith, and radial reflector.

3D model (shown in Fig. 6 (c)): employs a three-dimensional model of the entire reactor in order to calculate the neutron cross-sections of each material zone, including fuel rod, coolant, monolith, radial reflector, and axial reflector.

#### 4.1.1. Mesh refinement effect

To investigate the effect of mesh refinement, the 3D pin-cell model at the nominal state point ( $T_{\text{fuel}} = 298 \text{ K}$ ,  $T_{\text{cool}} = 298 \text{ K}$ ) is used as an illustration. Fig. 7 depicts the horizontal cut of finite element mesh of the pin-cell for various Mesh Size Factor (MSF). The mesh size factor set at Gmsh serves as a global scaling factor for all mesh sizes to quantify the level of mesh refinement. When the number of MSF changes from 0.5 to 0.1, the number of finite elements increases.

To compare the geometric description capability of the finite element mesh utilizing different MSF, the relative volumetric errors for each material zone are displayed in Fig. 8. As illustrated, decreasing the MSF results in a finite element mesh that more accurately represents the actual core geometry. When MSF equals 0.5, the relative volumetric discrepancy between the finite element mesh and the actual geometry approaches 4.51%. Changing the MSF to 0.1 decreases the relative error to 0.73%. In addition, the coolant zone has the largest relative volume inaccuracy of all material zones, due to the fact that the diameter of the coolant channels is 0.9 cm, making a fine mesh generation rather difficult.

As stated in section 2.1.3, the SPH method requires the iteration between reference heterogeneous code and homogeneous code utilizing volumetric-integrated neutron fluxes. The reference heterogeneous results are obtained using OpenMC by explicitly describing the geometry of the core. However, the diffusion computation is performed using a mesh representation of finite elements. The

volume mismatch between the finite element mesh and the actual geometry prohibits the SPH factors from converging to the correct values. Fig. 9 depicts the  $k_{\text{eff}}$  convergence utilizing several MSF. Different MSFs display a comparable convergence trend throughout the SPH iteration process. However, the smaller the MSF, the closer the SPH iteration approaches the reference  $k_{\text{eff}}$  value as convergence approaches. With a relative error of 26.6 pcm, the  $k_{\text{eff}}$  at the final step is 1.46103 when MSF equals 0.1.

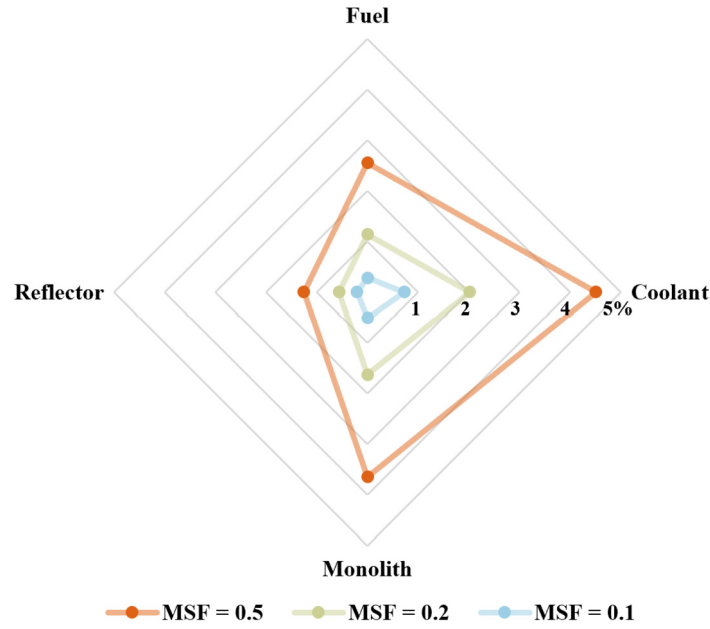
Fig. 10 depicts the distribution of neutron flux using multiple MSFs. It can be observed that reducing the mesh size factor (MSF) does not cause noticeable variation in the thermal and fast neutron flux distributions. Refining the meshes does not change the shapes of the neutron flux distributions, although slight variations in the maximum fast neutron flux can be observed. Specifically, the fast neutron flux varies between 14.88~14.92 with MSF=0.5 and between 14.95~14.99 with MSF=0.1, as depicted in Fig. 10.

#### 4.1.2. 2D core correction effect

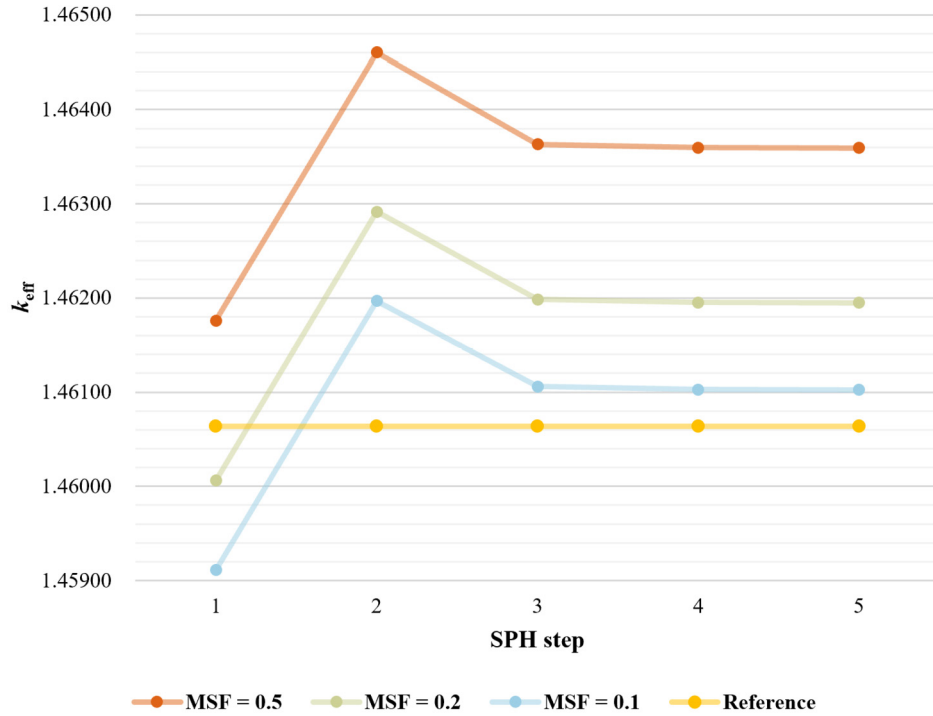
As mentioned in Section 2.1.3, the principle of SPH method is to preserve the reaction rate between the reference heterogeneous calculation and the homogeneous calculation. In this section we compare the fuel rod power errors using different calculation methods in order to evaluate the significance of SPH approach. The 2D model at the nominal state point ( $T_{\text{fuel}} = 298 \text{ K}$ ,  $T_{\text{cool}} = 298 \text{ K}$ ) is used as an example.

Fig. 11 illustrates the relative differences of fuel rod power at various states. First of all, when no correction method is applied, the homogeneous diffusion calculation completely miscalculated the power distribution in this 2D model, indicating that results from the direct diffusion calculation is not reliable, which is not surprising given the high level of heterogeneity in this model. The maximum relative error of power is about 18.7%, locating at the central fuel rod. With the incorporation of the SPH approach, computations may recover the OpenMC assembly powers regardless of whether the (n, xn) correction is accounted for, with the maximum





**Fig. 8.** Volumetric relative errors of various material zones using different mesh size factors (volumes calculated with the finite element mesh versus those calculated with the actual geometry).



**Fig. 9.** Comparison of the  $k_{eff}$  convergence trends using different mesh size factors.

and average relative error of power dropping to 1.5% and 0.5%, respectively.

Although it appears that the (n, xn) correction method does not contribute to the accuracy of the fuel rod power, it plays a crucial role in the correction of  $k_{eff}$ . As depicted in Table 3, the calculation with no cross-section correction yields the  $k_{eff}$  with a relative error of  $-4832$  pcm. Using the SPH method alone, the relative error of  $k_{eff}$  increases to  $-7471$  pcm. This is because that the neutron multiplication effects of (n, xn) reactions will break the neutron balance when OpenMC generates few-group cross-sections, causing a mismatch between the computed  $k_{eff}$  and the OpenMC reference  $k_{eff}$ . Only by combining the (n, xn) reaction correction and SPH

method can the calculation yield an accurate  $k_{eff}$  with a relative error of 27 pcm. Therefore, the absence of (n, xn) reaction correction could result in a pronounced relative error of  $k_{eff}$ , which should not be overlooked when using the SPH method.

#### 4.1.3. 3D core correction verification

To achieve precise forecasts for a nuclear reactor, a high-fidelity description of all fuel rods and holes is employed. Among the numerous approaches for generating cross-sections for 3D neutronic simulations, the most precise one is the 3D model. Fig. 12 depicts the fuel rod power error distribution using the 3D model (OpenMC v.s. FEMAS) at the nominal state point ( $T_{fuel} = 298$  K,

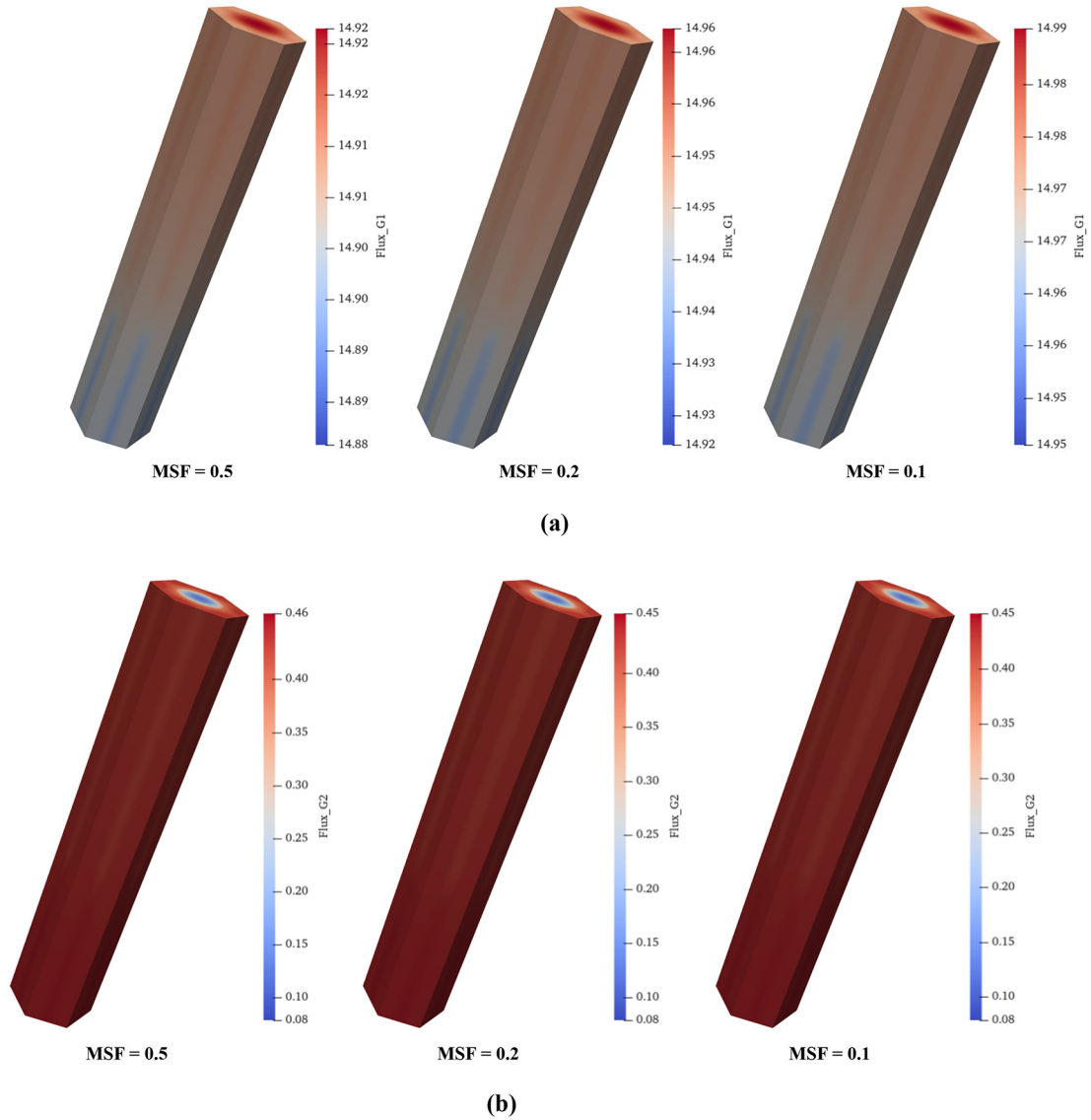


Fig. 10. Comparison of the neutron flux distribution using different mesh size factors.

**Table 3**  
Relative error of  $k_{\text{eff}}$  between OpenMC and FEMAS of 2D model.

| Calculation mode | Relative error of $k_{\text{eff}}$ |
|------------------|------------------------------------|
| None             | -4832                              |
| (n, xn)          | 2932                               |
| SPH              | 7471                               |
| (n, xn) + SPH    | 27                                 |

$T_{\text{cool}} = 298$  K). Without the introduction of (n, xn) reaction correction and SPH method, the radial distribution of power is poorly predicted with a maximum relative error of 6.4% and an average relative error of 4.5%. At the center of the core, the power is overestimated, and the error decreases gradually towards the periphery of the core. By utilizing the (n, xn) reaction correction and the SPH method, the power distribution between OpenMC and FEMAS agrees well, with a maximum relative error of 0.3% and an average relative error of 0.1%. In addition, in comparison to the  $k_{\text{eff}}$  of OpenMC ( $1.62004 \pm 0.00009$ ), the  $k_{\text{eff}}$  of FEMAS changes from 1.70212(5066 pcm) to 1.62016(7.2 pcm) with the adoption of the (n, xn) reaction correction and the SPH method.

## 4.2. Multi-physics coupling analysis

### 4.2.1. Comparisons of various coupling modes

Three calculation modes with differing degrees of coupling are studied to compare the differences introduced by the various models in coupling calculations. The specifics of these three modes are outlined below.

**N mode:** only calculations involving neutron diffusion calculation are performed. In this instance, only the neutron diffusion model is employed, the temperatures of all fuels, coolant channels, monolith and reflector are held constant, and the thermal expansion effect is disregarded.

**N-T mode:** the heat transfer calculation using the full core model is added to the N mode to account for the heat conduction effect. Consequently, the temperature field of the core is produced, and the effective temperature of all the materials can be updated, thereby modifying the cross-sections used in the calculation. This mode requires the iteration between the neutron diffusion calculation and the thermal conduction calculation.

**N-T-E mode:** the neutron diffusion effect, the thermal conduction effect and thermo-mechanics effect are all considered. Taking into account the thermo-mechanics model, the monolithic dis-

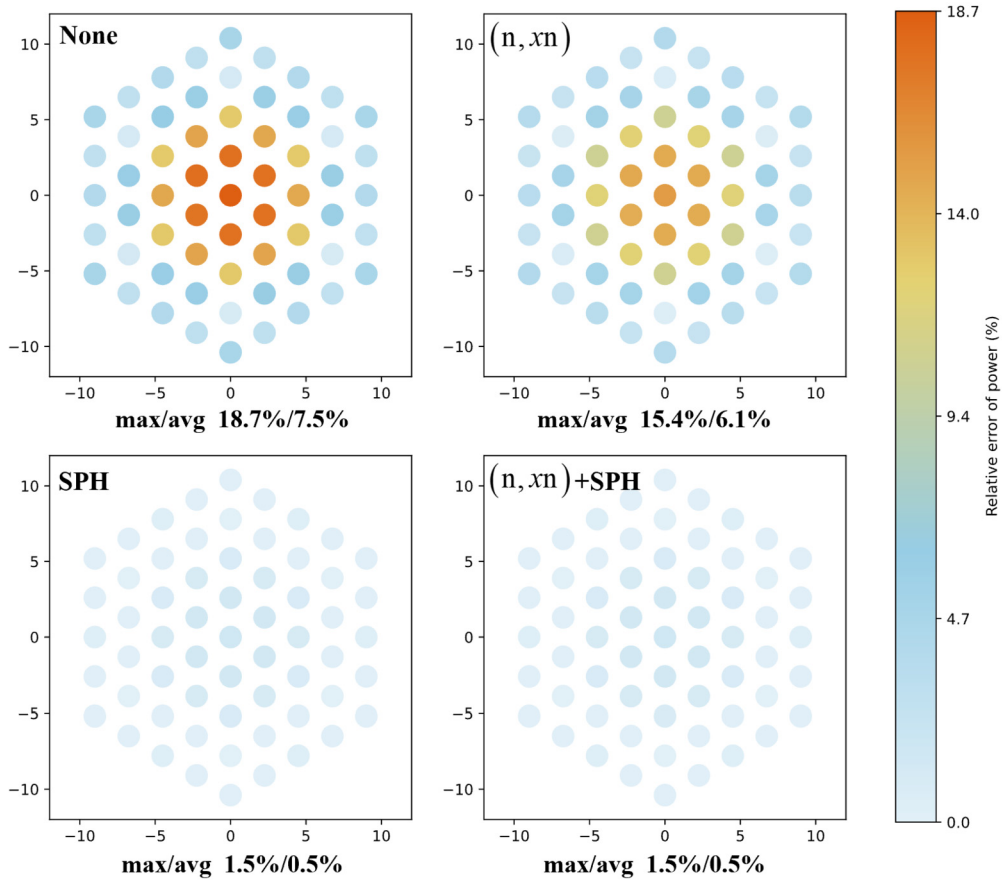


Fig. 11. Fuel rod relative power error (OpenMC v.s. FEMAS) distribution using the 2D model.

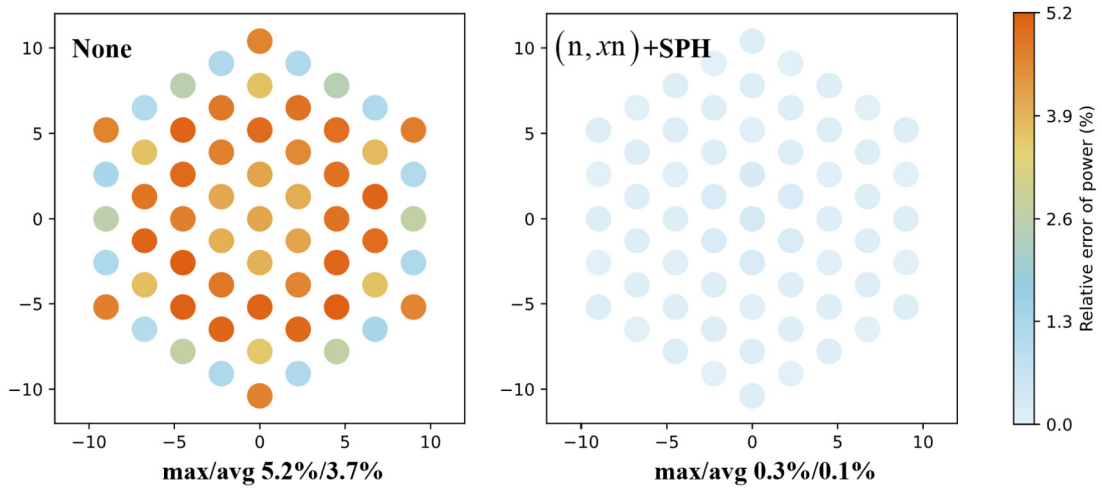


Fig. 12. Fuel rod relative power error (OpenMC v.s. FEMAS) distribution using the 3D model.

placement field and stress field are obtained, and the temperature field of entire reactor is modified. According to the displacement field, the location of each fuel and coolant channel alters. In conjunction with the 3D model, the geometry of the monolith and the average density of the monolith are updated.

As depicted in Fig. 13, based on the neutron diffusion calculation, the introduction of the thermal conduction calculation and the thermo-mechanics calculation will result in an increase in  $k_{eff}$  and a decrease in the power peaking factor. Comparing the N mode to the N-T mode, the Doppler effect provides reactivity feedback of 4361 pcm in  $k_{eff}$ . Additional integration of the thermo-mechanics

model (N-T-E mode) enhances the reactivity feedback by an additional 99 pcm. Unlike other solid-core reactors [23][24], the reactivity feedback of SIMONS is derived more from the Doppler effect than from thermal expansion effect which only accounts for 2.2% of total reactivity feedback. The reason is two-fold: first, the thermal-to-epithermal neutron spectrum of SIMONS results in a pronounced Doppler reactivity feedback effect; second, the thermal expansion coefficient of the graphite monolith is not as large as that for monoliths used in other solid-core reactors (such as the stainless-steel used in MegaPower), so the expansion effect is not as significant. In addition, the change curve of power peaking

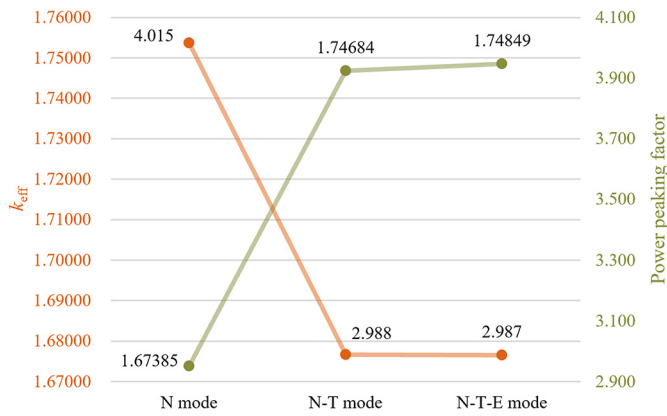


Fig. 13. Comparison of  $k_{\text{eff}}$  and power peaking factor using different calculation modes.

factor demonstrates that with the introduction of thermal conduction and thermo-mechanics calculations, the power distribution within the reactor core becomes more uniform. The Doppler effect reduces the power peaking factor from 4.015 to 2.988, and the thermal expansion impact increases it to 2.987.

To further investigate the effect of thermal conduction and thermo-mechanics, Fig. 14 shows the comparison of neutron flux distribution using different calculation modes. Compared to the thermal neutron flux, the fast neutron flux distribution demonstrates a more significant change when more physical fields are included. The Doppler effect results in a more uniform distribution of neutron flux, which is amplified by structural expansion.

As expected, the uniform change trend of neutron flux distribution will result in a flatter power distribution, as depicted in Fig. 15. The value of power indicates the effective power of each fuel rod, which are calculated using volume integration of power. Compared with the power distribution obtained by N mode, which varies from 906 W to 8063 W, and that of N-T-E mode varies from 923 W to 7678 W.

The temperature distributions obtained by N-T mode and N-T-E mode are depicted in Fig. 16. The outermost fuel rods have the highest temperatures, as demonstrated by the results. This is because, as indicated in Fig. 14, the thermalization of neutrons in the reflector produces a high number of thermal neutrons, creating a greater fission power in the fuel rods positioned on the reactor core's periphery. In addition, there are fewer coolant channels surrounding the locations of the hot spots compared to other fuel locations. The difference in temperature between the monolith and the reflector is minimal. Monolith has an effective temperature of 1009.7 K, while radial and axial reflectors have effective temperatures of 1011.4 K and 1007.7 K, respectively. The temperature distribution by N-T mode varies from 999.06 K to 1423.45 K and that by N-T-E mode varies from 999.18 K to 1353.86 K, with the constant boundary conditions for the wall of coolant channels, which may indicate the phenomenon of more uniform temperature distribution.

The effective temperature distribution of each fuel rod is depicted in Fig. 17. The left side of the diagram illustrates the numbering of each fuel rod in the core, while the right side depicts the effective temperature of the fuel rod in the N-T mode and N-T-E mode corresponding to different numbers. As shown by the temperatures calculated using N-T mode and N-T-E mode, the effect of thermal conduction and thermal expansion results in a more uniform fuel rod temperature distribution. When introducing the mean square error (MSE) to quantify the temperature distribution, the N-T mode yields an MSE value of 50.7, whereas the N-T-E mode yields an MSE value of 47.7. This further validates the aforementioned conclusion.

#### 4.2.2. Multi-physics coupling results

In the coupling calculation, the maximum number of iterations is set to 10. At step # 6, the relative error of  $k_{\text{eff}}$  falls below the pre-set convergence tolerances. At this point, the relative errors in  $k_{\text{eff}}$ , the radius of monolith and effective temperature of each material region between two successive iterations are  $5.15 \times 10^{-6}$ ,  $2.28 \times 10^{-6}$  and  $4.5 \times 10^{-5}$ , respectively. Therefore, it can be decided that the multi-physics simulation really converges. For clarity, the convergence trends of  $k_{\text{eff}}$  and radius of monolith for the multi-physics calculation are depicted in Fig. 18. At the first step, the  $k_{\text{eff}}$  is calculated by direct application of the OpenMC-generated cross-sections to the diffusion computation, yielding a value of 1.67385. When the coupling calculation converges at the sixth step,  $k_{\text{eff}}$  increases to 1.74849 with a relative rise of 4459.6 pcm, indicating that the adoption of thermo-mechanics calculation results in a substantial reactivity feedback. This positive reactivity feedback is not physically secure for the reactor, as is well understood. However, the SIMONS model utilized in this research is a simplified version with neutronic boundary conditions that are reflective. When a full-scale reactor with a vacuum boundary condition is simulated, it is expected that neutron leakage will cause the overall reactivity coefficient to be negative. In this paper, we will concentrate on demonstrating the multi-physics coupling method and leave a full-scale SIMONS analysis for future study.

Regarding the geometric change of the monolith, the radius increases from 13 cm to 13.075 cm with thermal expansion, and the height increases from 10 cm to 10.058 cm. As a result, the density of the monolith decreases to 97.6% of its initial value. Fig. 19 depicts the structural displacement of the monolith from the cold state to the hot state with a scaling factor of 5. Due to smaller displacement values, the coordinate changes of fuel rods and coolant channels inside the reactor core are less than those of fuel rods or coolant channels at the reactor core's edge.

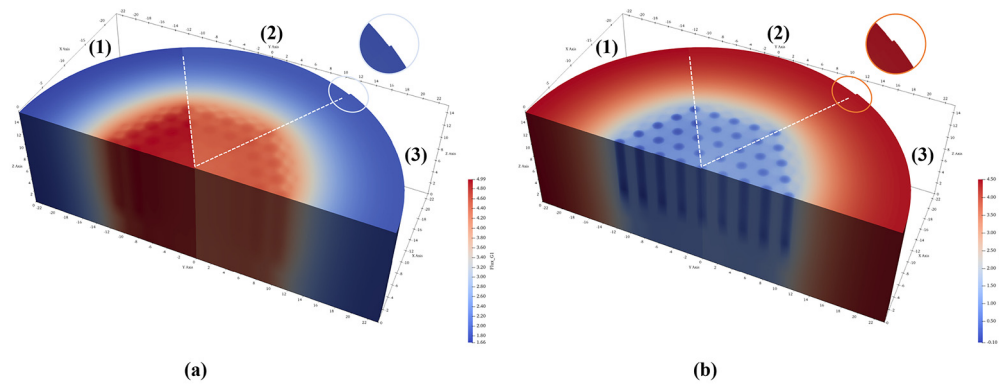
The results of various fields at reactor middle plane ( $H = 15$  cm) using neutronics and thermo-mechanics coupling calculations are displayed in the Fig. 20. This figure provides a more intuitive representation of the relationship between the various physical fields.

The two-group neutron flux distributions and the normalized power distribution are depicted in Fig. 21. As the temperature increases, the up-scattering effect of beryllium reflectors becomes more prominent, causing fast neutrons to accumulate in the reflector zone. In the meantime, thermal neutrons are present at the active core's perimeter due to the beryllium reflectors' significant thermalization effect. The interaction between the fast neutron and thermal neutron distributions results in a power distribution with a greater fraction at the active core's periphery. The power peaking factor of the entire core is 2.987 as a result.

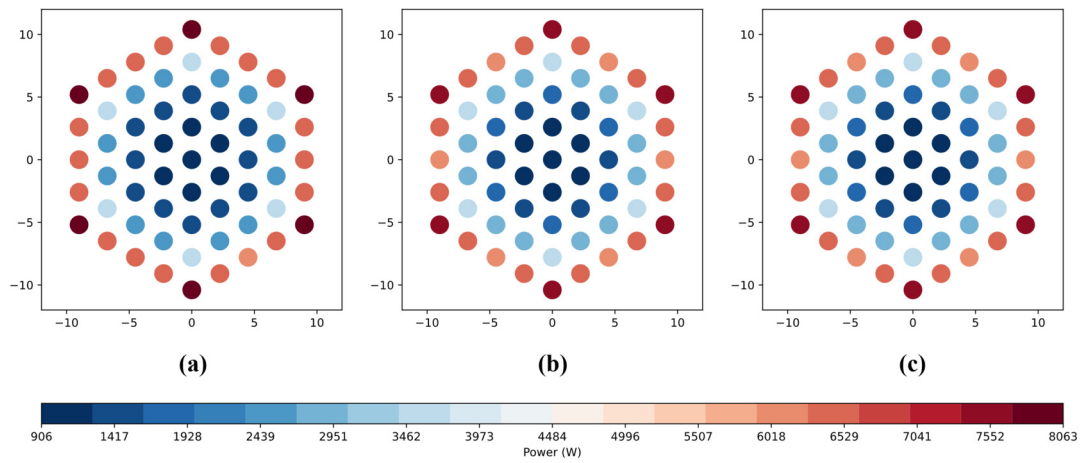
The polynomial fitting of the axial power distribution of two sample fuel rods is depicted in Fig. 22. Referring to the fuel configuration represented in Fig. 17, "Fuel 1" denotes the fuel rod at the core's center, whereas "Fuel 61" denotes the fuel rod at the core's periphery. The power distribution at both ends of each fuel rod is convex due to the moderation and reflection of neutrons by the beryllium reflector, which generates more thermal neutrons at the bottom and top of the fuel rod. Due to the fact that the radial reflector also contains beryllium, resulting in a greater population of thermal neutrons at the active core's periphery, the fuel rod at the active core's periphery has significantly more power than the central fuel rod.

Based on the previously stated core power distribution, it is possible to calculate the fuel zone's heat release rate. As indicated in Fig. 23, the temperature and displacement magnitude distribution of the simplified SIMONS may be determined utilizing thermo-mechanics. The largest temperature difference throughout the entire reactor is 354.68 K. The core expands from the cen-

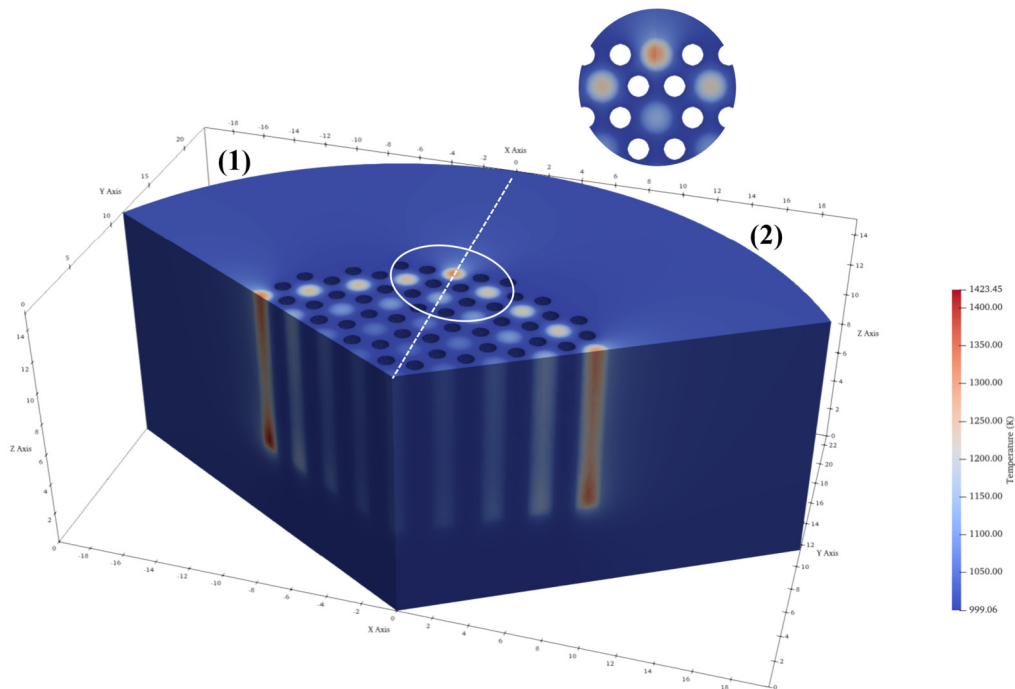




**Fig. 14.** Comparison of neutron flux distribution using different calculation modes. (1, 2 and 3 represent N mode, N-T mode and N-T-E mode respectively; a and b represent neutron flux of group 1 and group 2 respectively.)



**Fig. 15.** Comparison of power distribution using different calculation modes. (a: N mode; b: N-T mode; c: N-T-E mode).



**Fig. 16.** Comparison of temperature distribution using different calculation modes. (1 and 2 represent N-T mode and N-T-E mode respectively; 3 represents the local temperatures of the edge fuel.)

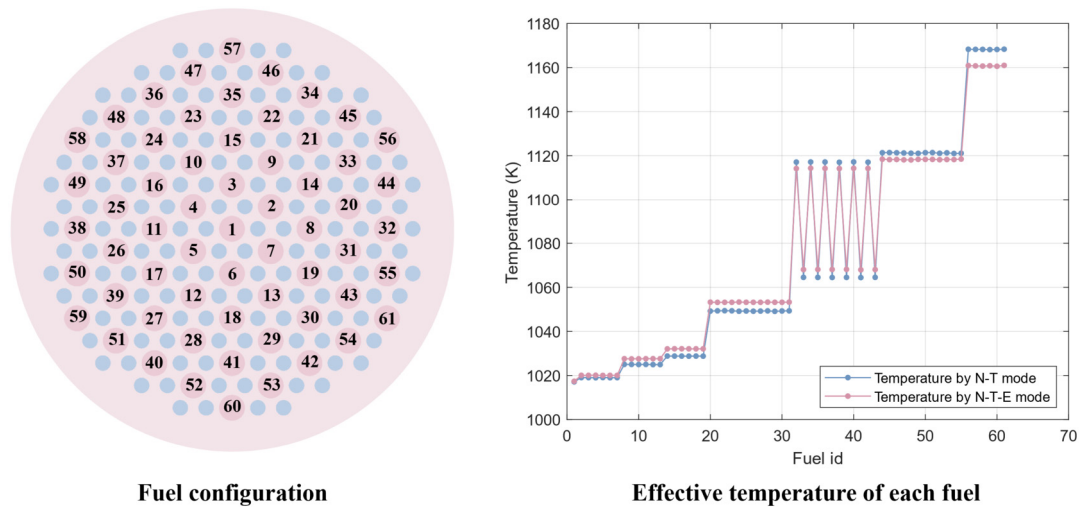


Fig. 17. Effective temperature of each fuel rod using different calculation modes.

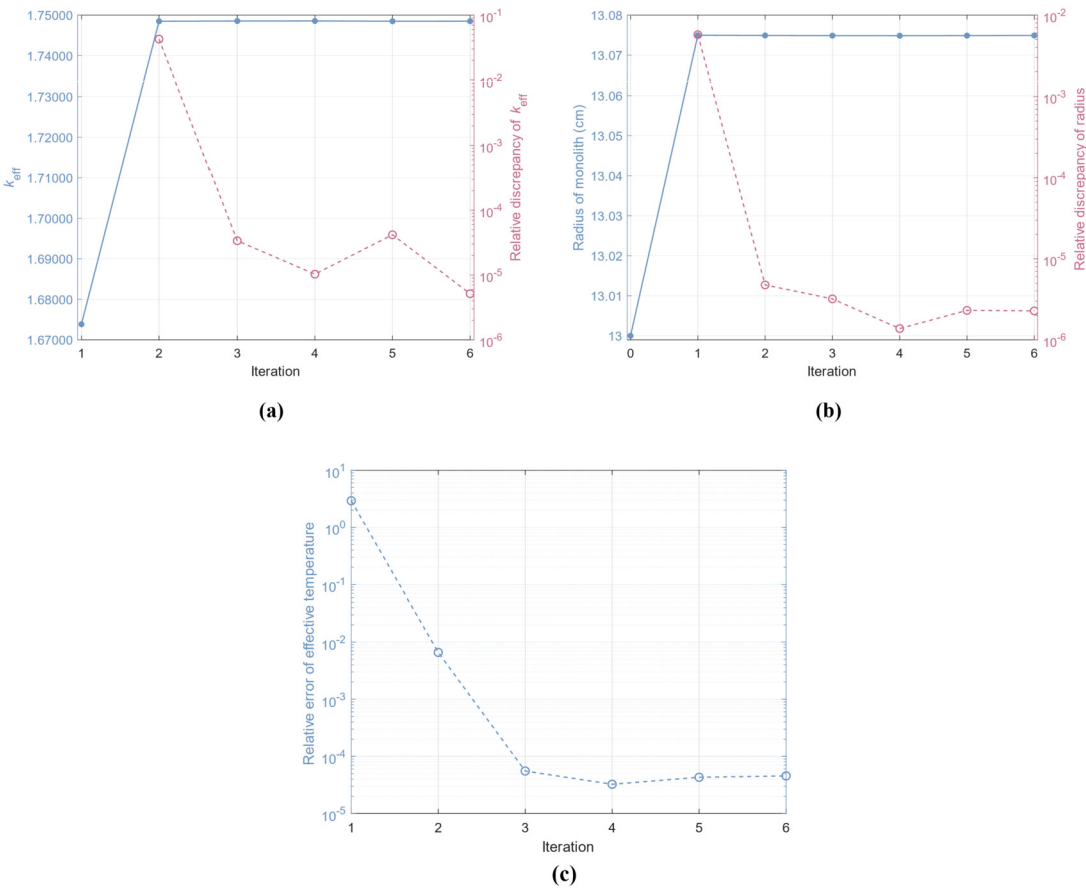


Fig. 18. Iteration convergence trends of  $k_{eff}$ , radius of monolith and effective temperature.

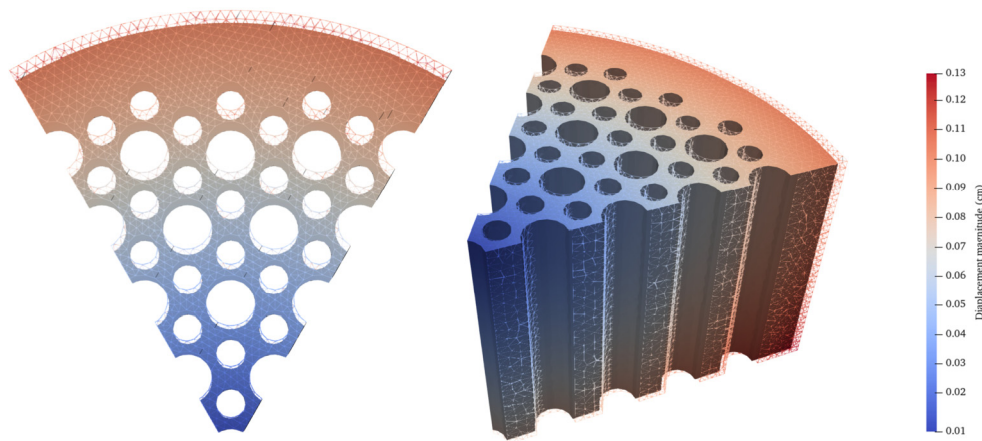


Fig. 19. Structural displacement of the monolith from the cold to the hot state (with the scaling factor of 5).

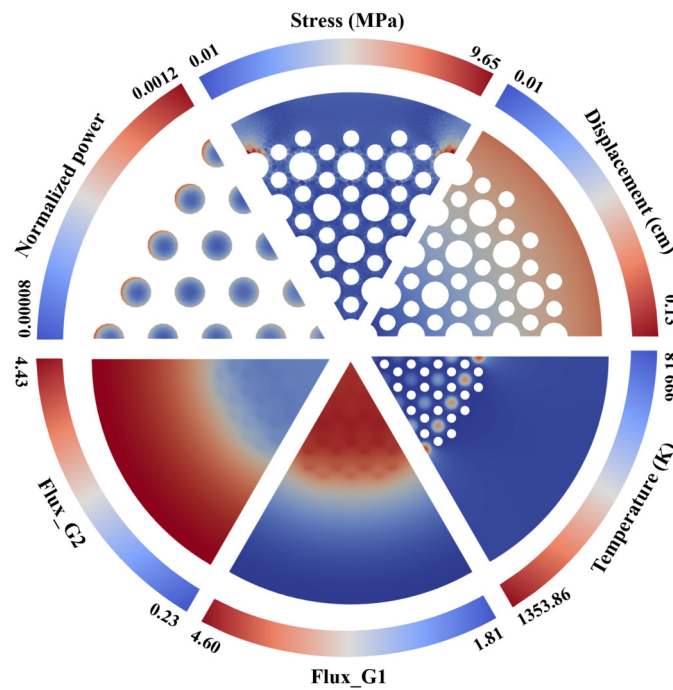


Fig. 20. Cross-sectional view of neutronics/thermo-mechanics coupling calculation ( $H=15$  cm).

ter to the periphery, as represented by the displacement field. As graphite's linear thermal expansion coefficient is relatively low, the thermal expansion impact is not as prominent as in other solid reactors. Nonetheless, it generates a series of desirable effects, including reactivity feedback, geometric expansion, and others. In addition, the high temperature of the entire reactor appears in the area of the fuel rods, and the temperature difference between the monolith and reflector is minimal. Due to the difference in their thermal expansion coefficients, they will eventually achieve radial separation in this instance.

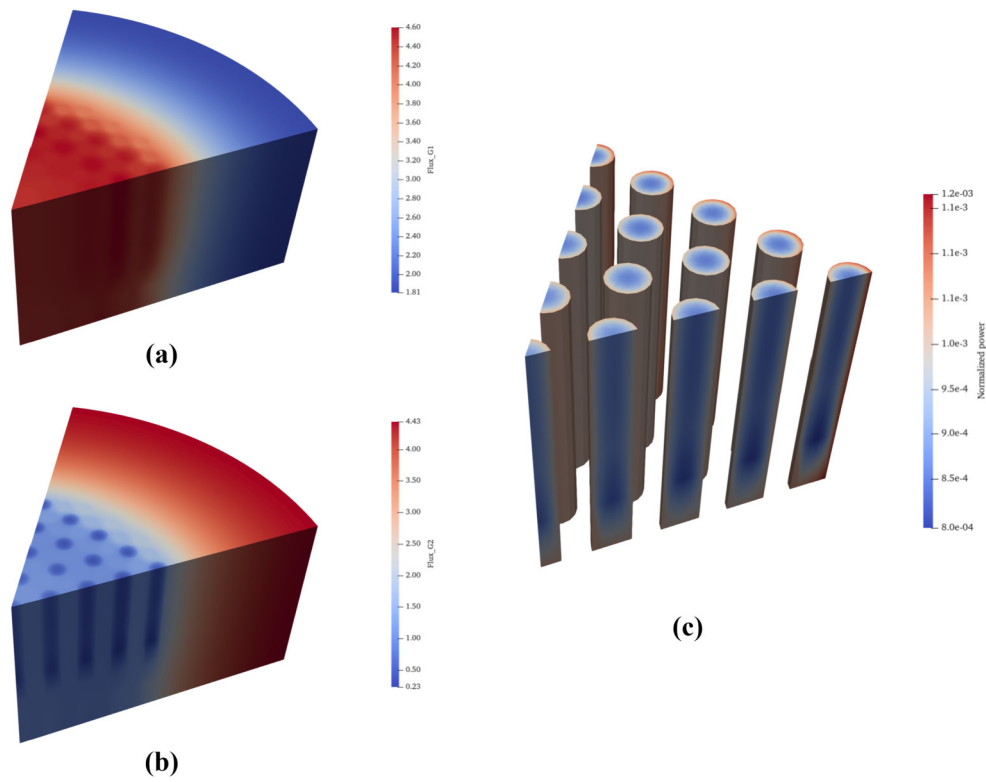
The Von Mises stress distribution of the simplified SIMONS is depicted in Fig. 24. At the wall in contact with the outermost fuel rods, where the temperature gradient develops dramatically and the thermal stress rises, the maximum Von Mises stress of the monolith occurs with a value of 9.65 MPa. This satisfies the specifications for the integrity of the entire monolith, given the ultimate tensile strength of graphite is 14 MPa. The maximum Von Mises stress of the reflector occurs at the part in contact with the monolith, where has a greater temperature gradient. The maximum Von

Mises stress value is 136.54 MPa, which is below the given the ultimate tensile strength of beryllium of 345 MPa.

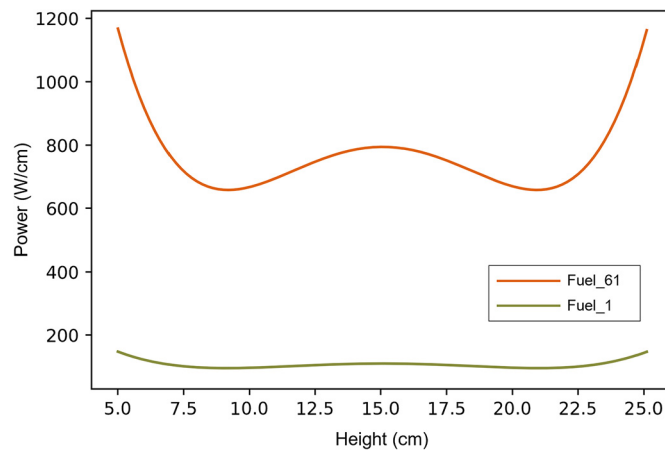
## 5. Conclusion

In this work, to fulfill the research and development requirements of innovative helium-xenon cooled mobile nuclear power systems, FEMAS, a 3D high-fidelity multi-physics coupling platform is developed. This platform integrates the open source codes OpenMC/deal.II/FEniCS to enables the prediction of multi-physics fields in micro reactors.

In this research, OpenMC is applied to construct the neutron cross-section library for finite element neutron diffusion calculations. To ensure neutron equilibrium and minimize homogenization errors, the SPH method and the  $(n, xn)$  correction are applied to OpenMC-generated cross-sections. With the adoption of these methods, it is discovered that the  $k_{eff}$  and power distribution calculation models are in good agreement with the reference solutions.



**Fig. 21.** Schematic views of the 3D neutronic results. (a) neutron flux of group 1; (b) neutron flux of group 2; (c) normalized fuel rod power.



**Fig. 22.** Polynomial fitting of axial power distribution of two sample fuel rods.

Simultaneously, based on a scaled model of the SIMONS micro reactor, this research performs a series of simulations and analyses, including from the OpenMC cross-section generation, reactor modeling, and multi-physics simulations. The results demonstrate that the coupling platform can accurately predict the fuel rod power distribution, the temperature distribution and the stress and strain distributions of SIMONS. Given 240-kW of thermal power, the local temperature difference of the entire reactor is 354.68 K, and the deformation rate caused by thermal stress is 2.4%, the maximum stress measures 9.65 MPa, and the reactivity feedback due to heat conduction and thermal expansion is 4459.6 pcm. It is demonstrated that the addition of thermal conduction and thermo-mechanics calculations, as opposed to neutronic computation alone, will introduce reactivity and geometric feedbacks that influence the reactor core's safety parameters.

During the era of high-precision computing, this platform may evaluate the performance of the core to guarantee that its design satisfies the specifications for ultra-long life and inherent safety. In contrast to standard coupling simulations, the thermo-mechanics model provides extra degrees of freedom, i.e., the stress and displacement distributions, which can be utilized to facilitate more strict and comprehensive analyses of the reactor. This research has paved the way for a more thorough multi-physics examination of SIMONS. Future study should foresee a full-scale reactor simulation with more realistic boundary conditions, such as vacuum boundary condition for neutronics and heat-transfer boundary condition for He-Xe coolant. In addition, depletion and transient analyses are in progress. It should also be noted that the current neutronic calculations are performed in serial, and require a substantial wall-clock time. Parallelization is under development in order to perform accurate simulations for more complicated problems.



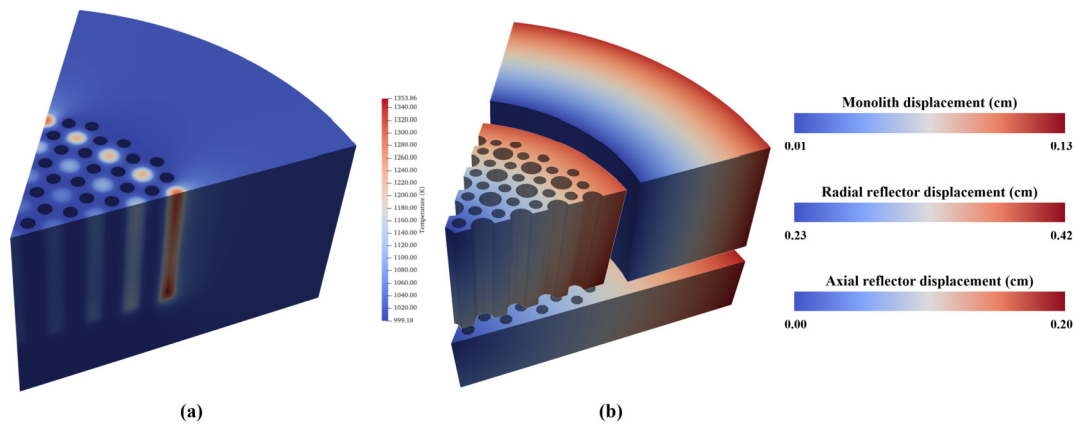


Fig. 23. Thermo-mechanics results. (a) the temperature distribution; (b) the displacement magnitude distribution.

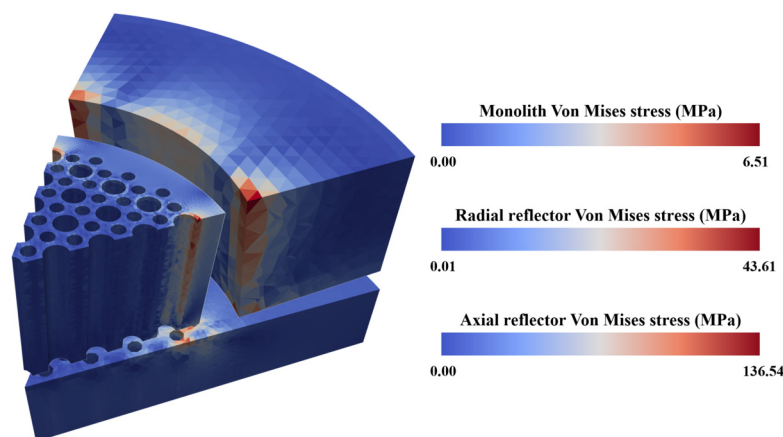


Fig. 24. The Von Mises stress distribution.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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